BTS-VII-11.17-1268 Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ]

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B.Tech. Degree VII Semester Examination November 2017

ME 1703 MACHINE DESIGN II

(2012 Scheme)

(Use of approved design data book is permitted and suitable data may assumed if necessary)

Time : 3 Hours | Maximum Marks : 100

PART A
(Answer *ALL* questions)

$(8 \times 5 = 40)$

I. (a) Derive the relationship for the length of the belt in terms of centre distances and diameters of the pulleys.

(b) Derive the equation of moments due to normal reaction, frictional force, and external force for an external contracting shoe brake. Write expressions for torque capacity and hinge pin reactions and show the same reactions hold good for internal expanding shoe brake also.

(c) Explain the procedure for designing a pair of bevel gears.

(d) Why is dynamic load induced in the gear teeth? Explain the procedure of designing for dynamic load using Buckingham equation.

(e) Compare rolling contact bearing with sliding contact bearing.

(f) Write a note on hydrodynamic theory of lubrication.

(g) How can you represent surface finish and tolerance in working drawings?

(h) What are the general design recommendations for rolled sections?

PART B

$(4 \times 15 = 60)$

II. A cone clutch is used to connect an electric motor running at 1440 rpm with a machine which is stationary. The machine is equivalent to a rotor of mass 150 kg and radius of gyration 250 mm. The machine has to be brought to the full speed of 1440 rpm from a stationary condition in 40 seconds. The semi cone angle alpha is 12.5 degrees. The mean radius of the clutch is twice the face width. The coefficient of friction is 0.2 and the normal intensity of pressure between the contacting surfaces should not exceed 0.1 N/mm^2. Assuming uniform wear condition, calculate:

(i) The inner and outer diameters.
(ii) The face width of friction lining.
(iii) The force required to engage the clutch.
(iv) The amount of heat generated during each engagement of the clutch.

OR

(P.T.O.)

III. It is required to design a leather cross belt drive to connect 7.5 kW, 1440 rpm electric motor to a compressor running at 480 rpm. The distance between the centre of the pulleys is twice the diameter of the bigger pulley. The belt should operate at 20 m/s approximately and its thickness is 5 mm. Density of leather = 950 kg/m^3 and ultimate tensile strength is 25 MPa.

IV. It is required to design a two stage spur gear reduction unit with 20 degree full depth involute teeth. The input shaft rotates at 1440 rpm and receives 10 kW power through a flexible coupling. The speed of the output shaft should be approximately 180 rpm. The gears are made of plain carbon steel 45C8 (ultimate tensile strength = 700 N/mm^2) and heat treated to a surface hardness of 240 BHN. The gears are to be machined to the requirements of grade 10. The service factor can be taken as 1.5.

(i) Assuming the dynamic load to be proportional to the pitch line velocity, estimate the required value of the module. The factor of safety is 1.5.
(ii) Select the first preference of the module and determine the correct value of factor of safety for bending.
(iii) Determine the factor of safety for pitting.

OR

V. Design a worm and a worm gear drive using double start threads with 50 teeth for a speed reduction by 25. Pinion (worm) rotates at 600 rpm and transmits 25 kW. Assume 20 degree full depth teeth.

VI. Design a journal bearing for 12 MW, 1000 rpm steam turbine, which is supported by two bearings. Take the atmospheric temperature as 16°C and operating temperature of oil as 60°C. Assume viscosity of oil as 23 centi Stokes, bearing material is babbit liner on steel or cast iron with factor of safety as 1.5.

OR

VII. A 30BC03(SKF 6306) deep groove ball bearing is to operate at 1600 rpm and carries a 8 kN radial load and 6 kN thrust load. The bearing is subjected to a light shock load. Determine the rating life of the bearing.

VIII. Explain with sketches, the points to be considered while designing parts manufactured by forging.

OR

IX. What are the general recommendations for welded parts? Explain with sketches.

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